P3S2 · WALL-MODAL

Membrane shear-wall cantilever — fundamental in-plane period vs exact Timoshenko beam

Membrane shear-wall cantilever — fundamental in-plane period converges to the exact Timoshenko beam

PASS

error −0.1913% · tol 0.5%

§2 TECHNICAL FEATURES TESTED

- Production **membrane wall** shell (STRIX engine)
- Self-weight mass through the production seismic-mass path
- Exact **Timoshenko cantilever** frequency equation (no discretisation)
- Shear coefficient κ = 10(1+ν)/(12+11ν) (Cowper)
- Four-level mesh refinement 3×6 → 24×48
- Reference independently checked by a converged 1-D Timoshenko FE

§3 PROBLEM DESCRIPTION

A shear wall is the stiffest lateral element of most buildings, and its in-plane period sets the spectral demand. This study checks that the production **membrane wall** of the STRIX engine reproduces the physics of a cantilever wall exactly as beam theory predicts. The wall is `3 m × 9 m`, `t = 200 mm`, fixed at the base and braced out-of-plane at every node (the slab/diaphragm idealisation); its only mass is its self-weight, lumped by the production seismic-mass rule on the **horizontal** degrees of freedom. Because the vertical degrees of freedom carry no mass, the wall cross-section has no rotary inertia, so the matching beam theory is the **Timoshenko cantilever without rotary inertia** (bending and shear flexibility, translational inertia only). Its exact fundamental frequency is the first root of the characteristic determinant built from `EIψ″ + κGA(w′ − ψ) = 0` and `κGA(w″ − ψ′) + ρAω²w = 0` with `w = ψ = 0` at the base and zero moment and shear at the top — Huang's cantilever frequency equation with the rotary-inertia term removed. With Cowper's `κ = 0.845` (ν = 0.2) it gives T_1 = `0.0505932 s`, about 6 % longer than the Euler-Bernoulli value — shear flexibility is not negligible at H/W = 3. The exact root was checked against a converged 1-D Timoshenko finite-element solution (agreement 7×10^{-7} %) and against the Euler limit when the shear stiffness is made rigid.

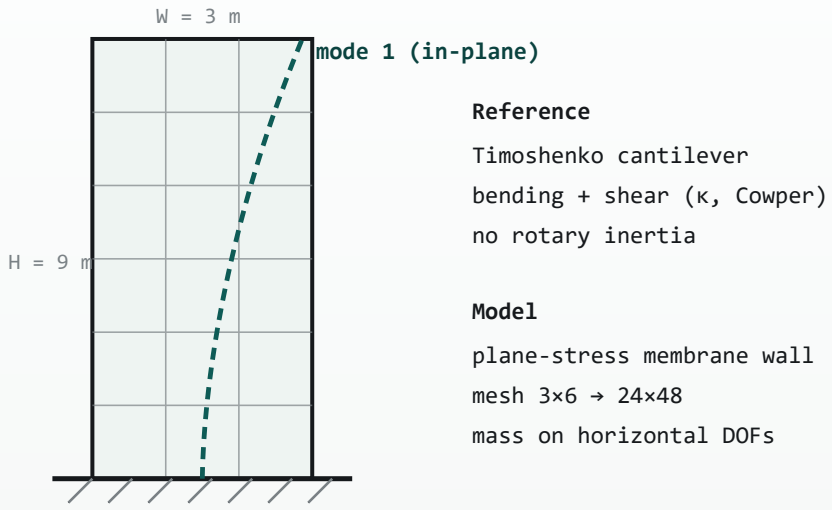


Figure P3-S2. Cantilever shear wall (3 m × 9 m, t = 200 mm) and its first in-plane mode; the plane-stress membrane model is refined from 3×6 to 24×48 elements and compared with the exact Timoshenko cantilever.

§4 GEOMETRY, PROPERTIES & LOADING

MODEL		MATERIAL		REFERENCE	
Wall W × H	3000 × 9000 mm	Model	Concrete · elastic	Theory	Timoshenko, no rotary inertia
Thickness t	200 mm	E	30000 MPa	κ (Cowper)	0.84507
Meshes	3×6, 6×12, 12×24, 24×48 (columns × rows)	ν	0.2	Analysis	Eigenvalue (modal)
Out-of-plane	restrained (diaphragm)	γ	2.4×10^{-5} N/mm³	Engine	STRIX engine (eigen)

§5 REFERENCE SOLUTION

Fundamental in-plane period — mesh 3×6: $2\pi/\omega_1$, $D(\omega_1) = 0$ = 0.050593 s

Fundamental in-plane period — mesh 6×12: $2\pi/\omega_1$, $D(\omega_1) = 0$ = 0.050593 s

Fundamental in-plane period — mesh 12×24: $2\pi/\omega_1$, $D(\omega_1) = 0$ = 0.050593 s

Fundamental in-plane period — mesh 24×48: $2\pi/\omega_1$, $D(\omega_1) = 0$ = 0.050593 s

Exact Timoshenko cantilever without rotary inertia: characteristic determinant from $EI\,\psi'' + \kappa GA(w' - \psi) = 0$, $\kappa GA(w'' - \psi') + \rho A\omega^2 w = 0$ with clamped–free boundary conditions (the frequency equation of T.C. Huang, J. Appl. Mech. 28 (1961) 579–584 with the rotary-inertia term removed); shear coefficient $\kappa = 10(1+\nu)/(12+11\nu)$ (G.R. Cowper, J. Appl. Mech. 33 (1966) 335–340, rectangle). Independently checked against a converged 1-D Timoshenko finite-element solution (verification/internal/p3s2_timoshenko_selfcheck.ts).

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Fundamental in-plane period — mesh 3×6 (s)	mode 1 (X 68%)	0.05047	0.050593	−0.2441%	PASS
Fundamental in-plane period — mesh 6×12 (s)	mode 1 (X 66%)	0.050456	0.050593	−0.2722%	PASS
Fundamental in-plane period — mesh 12×24 (s)	mode 1 (X 65%)	0.050481	0.050593	−0.2217%	PASS
Fundamental in-plane period — mesh 24×48 (s)	mode 1 (X 64%)	0.050496	0.050593	−0.1913%	PASS

§7 MESH CONVERGENCE OF T_1

MESH	T_1 (S)	DIFF
3×6	0.05047	−0.2441%
6×12	0.050456	−0.2722%
12×24	0.050481	−0.2217%
24×48	0.050496	−0.1913%

The fundamental in-plane period approaches the exact value as the mesh is refined and settles about `−0.19 % at 24×48`. The remaining gap is a **definition difference**, not a discretisation error: the wall is a plane-stress continuum whose fixed base also restrains in-plane warping, while beam theory represents shear through the single coefficient κ; both effects are second order at H/W = 3. The tolerance (0.5 %) was fixed only after this study, not before it.

§8 CONCLUSION

PASS

The STRIX engine membrane wall reproduces the exact Timoshenko cantilever period at every mesh, converging to within two tenths of a percent of the exact value; the residual is the continuum-versus-beam definition gap. In-plane bending and shear stiffness of the production wall element, and the production self-weight mass path, are therefore consistent with classical beam theory. (The former edition of this card, which swept the artificial stabilisers of a third-party shell element, does not apply to the STRIX engine shell, which has no such parameters.)

§9 REFERENCES & ANALYSIS FILES

1. T.C. Huang (1961) — *The effect of rotatory inertia and of shear deformation on the frequency and normal mode equations of uniform beams with simple end conditions*, J. Appl. Mech. 28, 579–584 (cantilever frequency equation; used here with the rotary-inertia term removed).
2. G.R. Cowper (1966) — *The shear coefficient in Timoshenko's beam theory*, J. Appl. Mech. 33, 335–340 (rectangle: κ = 10(1+ν)/(12+11ν)).
3. STRIX headless harness: `buildEigenPy → STRIX engine (eigen route) → modal_props.txt`; exact root and its 1-D FE check in `P3S2_timoshenkoWall.ts` / `p3s2_timoshenko_selfcheck.ts`.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · P3S2/` `sha256 (pending publish)`